AI Programming Foundations Capstone Project: Analyzing Airbnb Listings in New York City

Marius Jochheim

Udacity

AI Programming with Python Nanodegree

August 4, 2026

Overview

This project implements a reproducible Python workflow for cleaning, exploring, and visualizing the New York City Airbnb Open Data dataset (Gomonov, 2019).The dataset is available from Kaggle at https://www.kaggle.com/datasets/dgomonov/new-york-city-airbnb-open-data. The analysis examines how listing prices and annual availability vary across New York City neighbourhood groups and room types. The complete workflow is contained in the accompanying `data_workflow.ipynb` notebook.

Dataset Description

The New York City Airbnb Open Data dataset contains listing-level information for Airbnb properties in New York City during 2019. The original CSV file contains 48,895 rows and 16 columns, with each row representing one listing.

The analysis focused primarily on `price`, `neighbourhood_group`, `room_type`, `availability_365`, `minimum_nights`, `number_of_reviews`, `reviews_per_month`, and `calculated_host_listings_count`. Latitude, longitude, listing identifiers, listing names, and host information were retained but were not central to the main comparisons.

The dataset includes the five neighbourhood groups Manhattan, Brooklyn, Queens, the Bronx, and Staten Island. Its room-type categories are entire home or apartment, private room, and shared room.

Workflow Description

The project followed a sequential workflow consisting of data ingestion, inspection, cleaning, exploratory analysis, visualization, and interpretation. This structure was chosen to make the analytical process transparent and repeatable. Danchev describes reproducibility as a principle that should be incorporated throughout the data-science lifecycle rather than treated only as a final documentation activity (Danchev, 2022).

Ingestion

The CSV file was stored within the repository and loaded with `pandas.read_csv()`. The notebook then inspected the dimensions, column names, data types, missing values, duplicated rows, categorical values, and numerical ranges before applying any transformations.

Cleaning

Two reusable cleaning functions were created. The first converted `last_review` to a datetime type and handled missing review-rate values. The second removed records that violated explicit validity rules, including non-positive prices.

Exploratory Analysis

A reusable exploratory-analysis function generated numerical summaries, neighbourhood-level summaries, room-type summaries, combined neighbourhood and room-type comparisons, and a Pearson correlation matrix. Additional analyses examined zero availability and compared mean and median prices before and after limiting prices to the 99th percentile.

Visualizations

Four visualizations were created:

1. median price by neighbourhood group and room type;
2. price distributions by room type;
3. median annual availability by neighbourhood group and room type;
4. correlations between numerical variables.

Summary

The final notebook section interpreted the calculated results, documented the main limitations and assumptions, and identified questions that could not be resolved from the available variables.

Key Decisions and Assumptions

Missing Review Information

The original dataset contained 10,052 missing values in both `last_review` and `reviews_per_month`. Inspection showed that these missing values were associated with listings that had zero recorded reviews.

Missing `reviews_per_month` values were therefore replaced with zero only where `number_of_reviews` was also zero. This preserved unreviewed listings while representing their observed monthly review activity numerically. Missing `last_review` values were retained because a listing with no reviews has no meaningful last-review date.

This variable-specific approach was preferable to deleting every incomplete record. Missing-data treatment can affect representativeness and introduce bias, so the meaning and pattern of missingness should be considered before selecting a treatment (Kang, 2013).

The small numbers of missing listing names and host names were retained because those descriptive fields were not needed for the principal price and availability analyses.

Invalid and Extreme Values

The cleaning process removed 11 records whose prices were zero. A non-positive nightly price was treated as invalid for the planned price analysis. This reduced the dataset from 48,895 to 48,884 listings.

Large positive prices and long minimum-stay requirements were not automatically removed. Although these observations are unusual, the dataset provides insufficient evidence to classify them as errors. Removing them without an explicit validity rule could exclude legitimate listings and alter the observed distribution.

The price distribution was strongly right-skewed. The full cleaned dataset had a mean price of $152.76 and a median price of $106.00. For listings priced at or below the 99th percentile of $799, the mean decreased to $137.58, whereas the median changed only to $105.00. Median prices were consequently used for the main grouped comparisons because they were less sensitive to extreme values.

Prices above the 99th percentile were excluded only from Figure 2 in the accompanying notebook to improve its readability. They remained in the cleaned dataset and in the other analyses.

Choice of Exploratory Comparisons

Neighbourhood group and room type were selected as the central categorical variables because they allowed direct comparison of location and accommodation category. Price and availability were grouped by both variables so that their combined patterns would not be hidden by separate aggregate summaries.

The correlation matrix was included to examine linear relationships among the numerical variables. Correlation coefficients were interpreted as descriptive associations and not as evidence of causation.

Results and Interpretation

Listing Composition

The cleaned dataset contains 48,884 listings. Manhattan accounts for 44.31% of the listings, while Brooklyn accounts for 41.11%. Queens represents 11.59%, the Bronx 2.23%, and Staten Island 0.76%.

Entire homes and apartments are the largest room-type category, accounting for 51.97% of listings. Private rooms represent 45.66%, while shared rooms account for only 2.37%.

Prices by Neighbourhood Group and Room Type

Manhattan has the highest overall median listing price at $150 per night, whereas the Bronx has the lowest at $65. Across room types, entire homes and apartments have a median price of $160, compared with $70 for private rooms and $45 for shared rooms.

Figure 1 shows that entire homes and apartments have the highest median price in every neighbourhood group. Manhattan has the highest median prices for all three categories: $191 for an entire home or apartment, $90 for a private room, and $69 for a shared room. Brooklyn has the second-highest median entire-property price at $145, followed by Queens at $120. Entire properties in both the Bronx and Staten Island have a median price of $100.

The price premium associated with renting an entire property is particularly large in Manhattan and Brooklyn. An additional notable result is that the median shared-room price in Manhattan is higher than the median private-room price in every other neighbourhood group.

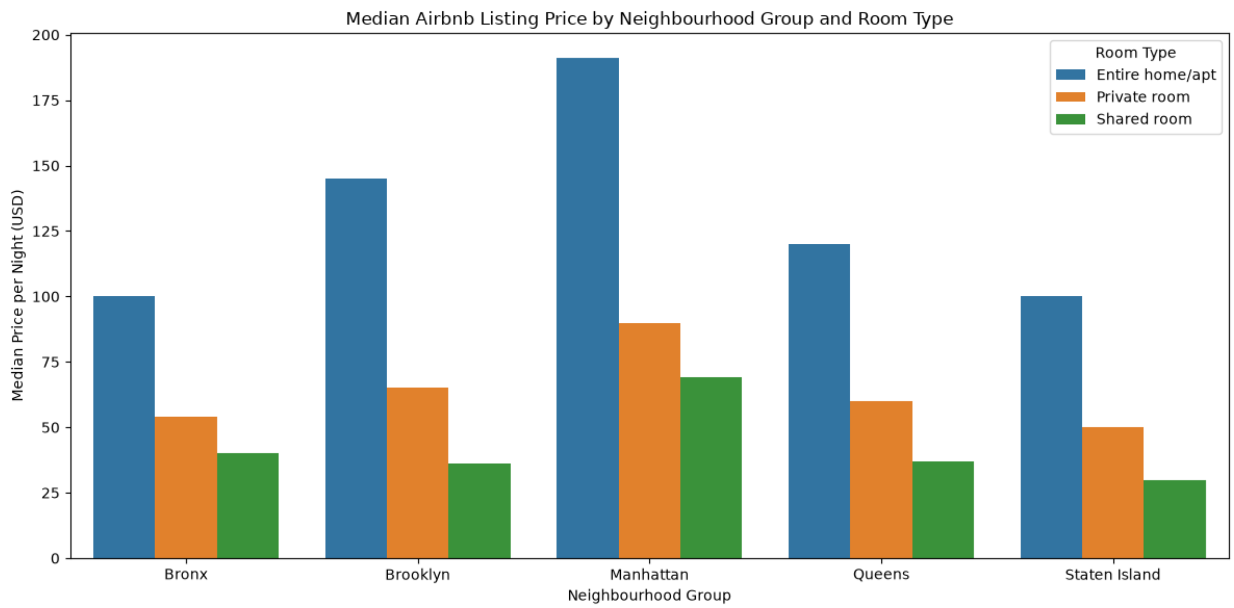

Figure 1

Median Airbnb listing price by neighbourhood group and room type.

Price Distributions

Figure 2 shows that the distributions differ clearly by room type. Entire homes and apartments have both the highest median and the widest interquartile range. Private and shared rooms are concentrated at lower prices, although both categories still contain relatively expensive listings.

All three distributions are right-skewed and contain many observations above their upper box-plot whiskers. These observations are statistical outliers under the box-plot convention but are not necessarily erroneous records. The figure therefore supports retaining the observations while using median values to describe typical prices.

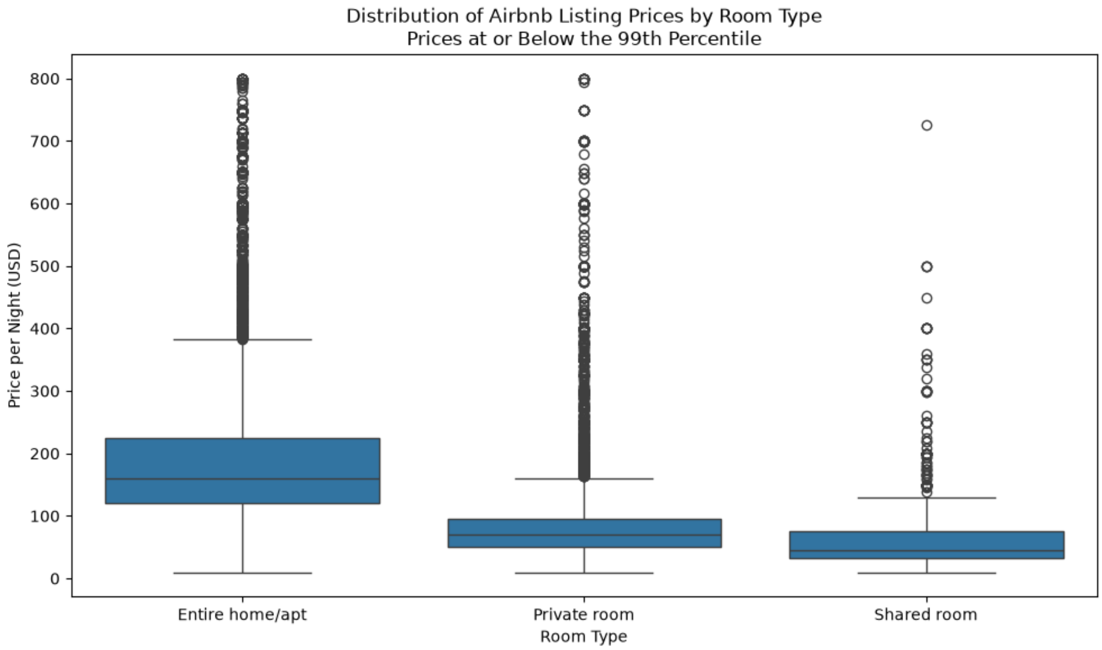

Figure 2
Distribution of listing prices by room type for prices at or below the 99th percentile.

Annual Availability

Figure 3 shows substantial variation in median annual availability. Staten Island has the highest typical availability for private rooms, at approximately 280 days, and for entire homes and apartments, at approximately 175 days. Manhattan and Brooklyn have much lower median availability for these categories, generally below 50 days.

Shared rooms show a different pattern. Their median availability is relatively high in Brooklyn and Queens but low in Staten Island. The results indicate that availability patterns depend on the combination of neighbourhood group and room type.

A total of 17,530 listings, or 35.86% of the cleaned dataset, have zero recorded availability. The proportion is highest in Brooklyn at 39.02%, followed by Manhattan at 37.40%. Staten Island has the lowest proportion at 11.26%.

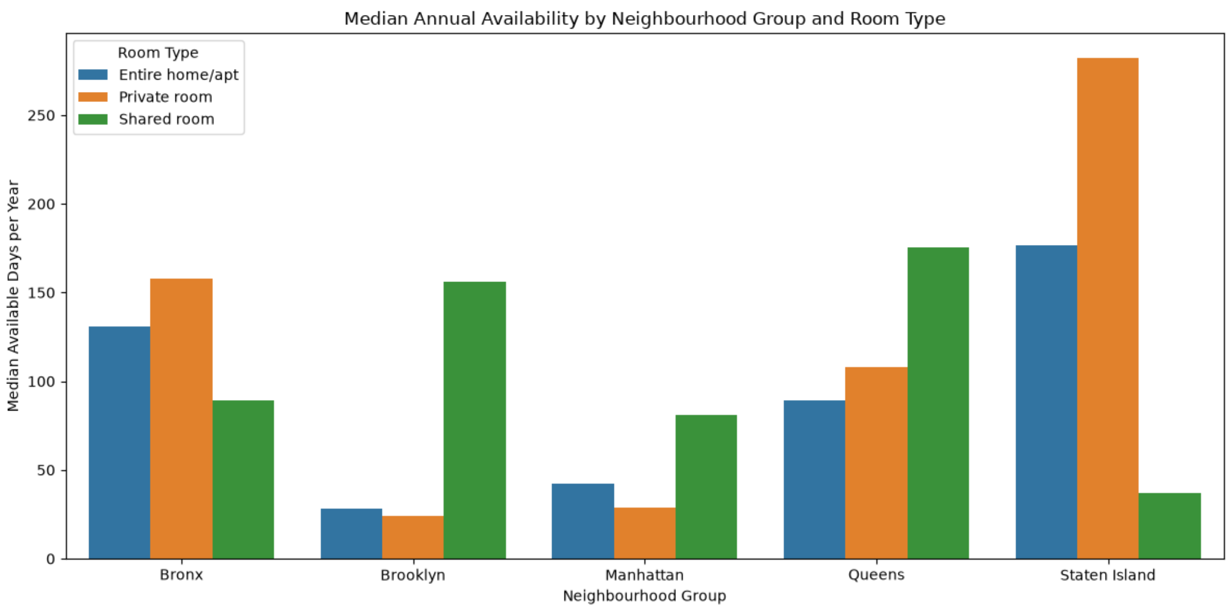

Figure 3
Median annual availability by neighbourhood group and room type.

Numerical Correlations

Most numerical variables have only weak linear relationships, as shown in Figure 4. The strongest correlation is between `number_of_reviews` and `reviews_per_month`, with a coefficient of 0.59. This moderate positive association is expected because both variables measure review activity.

A weaker but more substantively interesting relationship occurs between `calculated_host_listings_co` and `availability_365`, with a coefficient of 0.23. Listings associated with hosts managing more properties tend to have slightly greater recorded availability, but this relationship is weak and cannot establish causation.

Price has very weak correlations with all included numerical variables. Its largest numerical correlation is only 0.08, with annual availability. The grouped results therefore suggest that neighbourhood group and room type are more informative for describing price differences than the numerical variables included in the correlation analysis.

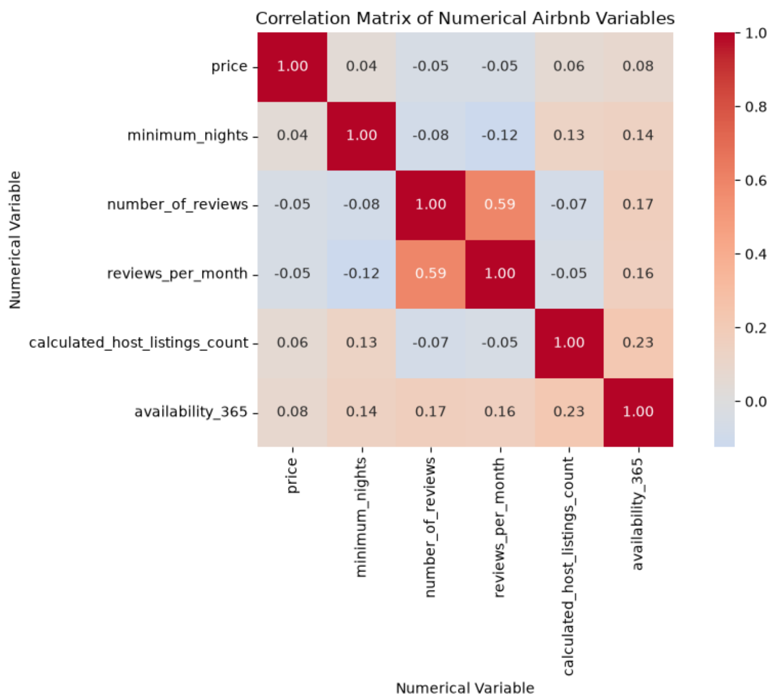

Figure 4
Pearson correlation matrix for selected numerical variables.

Responsible Practice: Bias and Data Quality

The findings are descriptive and should not be interpreted as causal. The dataset does not include several characteristics that may influence price, including property size, bedroom count, amenities, property condition, proximity to attractions, and public-transport access. Price differences associated with neighbourhood group or room type may therefore partly reflect unobserved variables.

The neighbourhood groups are also unevenly represented. Manhattan and Brooklyn together account for more than 85% of the cleaned observations, while Staten Island accounts for less than 1%. Aggregate findings are consequently dominated by Manhattan and Brooklyn, and conclusions about smaller groups are based on much smaller samples. The shared-room category is also limited, particularly in Staten Island, where only nine shared-room listings are present.

The `availability_365` variable is ambiguous. Low availability could indicate frequent bookings, but it could also result from host-blocked dates, seasonal operation, or an inactive listing. High availability could indicate lower demand, greater supply, or a newly created listing. Availability was therefore not treated as a direct measure of occupancy or demand.

Open datasets should not automatically be assumed to represent verified ground truth. Alsudais documents systemic collection errors and reproducibility problems across releases of the Inside Airbnb dataset (Alsudais, 2020). Although the present dataset was obtained from Kaggle rather than directly from an Inside Airbnb release, the same general caution applies to third-party platform data. The report therefore identifies the exact dataset source, retains the original CSV, documents all cleaning rules, and limits its conclusions to the supplied snapshot.

Future work could reduce these risks by validating selected listings against another source, documenting a precise download date or version, examining individual neighbourhood sample sizes, and repeating the analysis with data from additional collection periods.

Reproducibility

The repository contains the original dataset, the complete Jupyter Notebook, the generated figures, this report, a `requirements.txt` file, and a short README with execution instructions. A reviewer can create a virtual environment, install the pinned dependencies with

```
pip install -r requirements.txt
```

and run all cells in `data_workflow.ipynb` from top to bottom.

The notebook uses relative file paths, reusable functions, explicit cleaning rules, and intermediate validation checks. The unmodified data are stored in `df`, while the cleaned data are stored separately in `df_clean`. This separation preserves the original input and makes the transformations easier to inspect.

Git was used to record progress through multiple commits. Development work was performed on an additional branch before being integrated with the main branch. The commit history documents the progression from repository setup through data ingestion, cleaning, exploratory analysis, visualization, interpretation, and reporting.

These practices support computational reproducibility by preserving the code, data, software dependencies, and development history needed to repeat the analysis (Danchev, 2022).

Conclusion

The analysis found clear differences in Airbnb listing prices across New York City neighbourhood groups and room types. Manhattan is the most expensive neighbourhood group, and entire homes and apartments are consistently the most expensive room type. Listing prices are strongly right-skewed, making median prices more representative of typical listings than mean prices.

Availability patterns are more complex and depend on the combination of neighbourhood group and room type. They cannot be interpreted directly as occupancy or demand because the dataset does not explain why dates were unavailable. Most numerical variables have weak correlations with price, suggesting that location, room type, and unrecorded property characteristics are likely to be important for explaining price differences.

Further analysis could compare individual neighbourhoods, examine geographic patterns using latitude and longitude, incorporate additional property characteristics, and use repeated data releases to investigate changes over time.

References

Alsudais, A. (2020). Incorrect data in the widely used inside airbnb dataset. *CoRR*, *abs/2007.03019*. https://arxiv.org/abs/2007.03019

Danchev, V. (2022). Reproducible data science with python: An open learning resource. *Journal of Open Source Education*, *5*(56), 156. https://doi.org/10.21105/jose.00156

Gomonov, D. (2019). *New york city airbnb open data* [Accessed 2026-08-04]. Kaggle. https://www.kaggle.com/datasets/dgomonov/new-york-city-airbnb-open-data

Kang, H. (2013). The prevention and handling of the missing data. *Korean J Anesthesiol*, *64*(5), 402–406. https://doi.org/10.4097/kjae.2013.64.5.402